

Aegis Security Analysis: Classical vs. Quantum Computing

Security Estimates for Encryption Technologies

Method	Classical Computer	Quantum Computer
RSA-2048 (current standard)	1 trillion years	8 hours with sufficient qubits
ECC-256 (current standard)	10 billion years	9 hours with sufficient qubits
Kyber-768 (used in Aegis)	2^{164} operations (millions of years)	2^{82} operations (millions of years)
SPHINCS+-128f (used in Aegis)	2^{128} operations (billions of years)	2^{64} operations (thousands of years)

Breaking Aegis Security

With Classical Computers:

- **Kyber-768:** Would require approximately 2^{164} operations, which is far beyond what's achievable with current or projected classical computing technology. Even if every computer on Earth worked together, it would take millions of years.
- **SPHINCS+:** Would require approximately 2^{128} operations for a hash function collision, which remains computationally infeasible with classical computers. This is comparable to or stronger than AES-128, which is still considered secure for classical computing.

With Quantum Computers:

- **Kyber-768:** Security drops to approximately 2^{82} operations due to Grover's algorithm, but this still represents thousands of years of computation even with advanced quantum computers.
- **SPHINCS+:** Security level reduces to approximately 2^{64} operations, which while significantly lower than its classical security, would still require thousands of years with projected quantum computing power.

Security Reduction Factors

Algorithm	Classical Attack	Quantum Attack	Security Reduction
RSA-2048	1 trillion years	8 hours	10^{15}
ECC-256	10 billion years	9 hours	10^{14}
Kyber-768	millions of years	millions of years	10^3

Algorithm	Classical Attack	Quantum Attack	Security Reduction
SPHINCS+-128f	billions of years	thousands of years	10^6

Practical Implications

- **Current cryptographic standards** (RSA, ECC) will be essentially **broken immediately** once sufficiently powerful quantum computers are developed
- **Aegis’s algorithms** (Kyber and SPHINCS+) would still require **thousands to millions of years** to break even with quantum computers
- This makes Aegis suitable for protecting data that requires long-term security, including:
 - Government classified information
 - Healthcare records
 - Financial systems
 - Critical infrastructure
 - Long-term digital archives

Key Size and Performance Trade-offs

Algorithm	Security Level	Key Size	Speed
RSA-2048	112 bits (classical)	2048 bits	Fast
ECC-256	128 bits (classical)	256 bits	Very fast
Kyber-768	164 bits (classical) / 82 bits (quantum)	1184 bytes	Fast
SPHINCS+-128f	128 bits (classical) / 64 bits (quantum)	32 bytes	Slower (especially for signing)

The Aegis implementation demonstrates that post-quantum security is achievable with reasonable performance trade-offs and remains secure against both classical and quantum computing attacks for the foreseeable future.